

# ANAS AHMED

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## EDUCATION

**B.S. in Computer Science** *National University of Computer and Emerging Sciences (FAST-NUCES) — Karachi, Pakistan*

- **Timeline:** August 2022 - June 2026 | **Cumulative CGPA:** 3.73 / 4.00
- **National Merit: 99.8th Percentile** (Top 0.2% nationwide) out of all IT students in the HEC & P@SHA National Skill Competency Test (Scored 82/100 across Data Structures, Algorithms, OS, AI, Networking and Programming).
- **Honors:** Six-time Dean's List of Honors awardee. One time Rector's List of Honors awardee.
- **Language Proficiency:** IELTS Academic — Overall Band 8.0 (CEFR C1 Mastery).
- **Relevant Coursework:** Deep Learning for Perception, Artificial Intelligence, Design & Analysis of Algorithms, Data Structures & Algorithms, Computer Vision, Operating Systems.

**International Summer School (Metaverse Development)** *Universiti Teknikal Malaysia Melaka (UTeM) — Malacca, Malaysia | July 2025 – August 2025*

- Focused on immersive technology frameworks, 3D environment logic, and cross-cultural academic collaboration.

## RESEARCH & CORE PROJECTS

**Transformer Health Indexing System (FYP)** | *Industrial Client: K-Electric*

- Spearheading an industrial-grade Computer Vision detection pipeline utilizing YOLOv8 and PyTorch to automatically detect structural vulnerabilities on power distribution systems.
- Interfaced the deep learning processing models into an asynchronous microservices platform using FastAPI engines and Next.js interfaces.
- Conducted model weight optimization audits to perfectly balance the runtime latency and mean Average Precision (mAP) variables.

**CrossViewNet: Aerial to Ground Scene Synthesis** | *Academic Research*

- Modernized a legacy CVPR framework into modern PyTorch 2.x and integrated an 8-stage U-Net Pix2Pix GAN to synthesize photorealistic 360° panoramas from top-down aerial imagery.
- Optimized training parameters via Automatic Mixed Precision (AMP) to train the full pipeline and handle multi-scale hypercolumns within a tight 4GB VRAM constraint.

**Gaze Detection AI System** | *Academic Research*

- Fine-tuned EfficientNetB0 classification architectures to process real-time structural facial directional coordinates at ~25 FPS on native CPU environments.  
Combined MediaPipe face landmark arrays to programmatically dynamically map regional regions of interest (ROI) amidst shadows and low exposure variables.

## PROFESSIONAL EXPERIENCE

**Traxccel** — Karachi, Pakistan *Full-Stack Developer* | April 2026 – Present (*Full-time*)

- Engineered cross-platform enterprise web applications, internal CRMs, and management portals using React.js and a modular ASP.NET Core backend architecture with RESTful APIs.
- Designed high-throughput data access pipelines and securely mapped complex relational schemas to dynamic frontend dashboards using Entity Framework Core and SQL Server.
- Architected high-performance, responsive user interfaces with Bootstrap CSS while integrating enterprise communication workflows via Microsoft Graph API, SMTP, and external microservices.

**ShahKings** — Remote *Full-Stack Developer* | January 2024 – April 2026 (*Part-time*)

- Developed and optimized scalable web ecosystems using .NET Core MVC architectures and native JavaScript.
- Implemented end-to-end data encryption and API tracking to securely integrate the HBL Payment Gateway for the "Khas Foundation" platform.
- Co-engineered "Wallez," an offline-first hybrid mobile application using React and Capacitor, optimizing local database synchronization and zero-latency performance.

**Techmile Solutions** — Karachi, Pakistan *Developer Intern* | June 2025 – August 2025

- Eliminated manual quality assurance bottlenecks by scripting automated testing suites via Playwright and Trigger.dev pipelines.
- Constructed Python-based web scrapers for structured data aggregation and contributed modules to production React Native builds.

**FAST-NUCES** — Karachi, Pakistan *Student Lab Assistant (Programming Fundamentals)* | September 2025 – January 2026

- Instructed and evaluated a class of 50+ undergraduate students on algorithms, data logic, structural debugging routines, and syntax execution in C along with GitHub basics.

## TECHNICAL SKILLS

- **AI & Machine Learning:** PyTorch, TensorFlow, YOLOv8, EfficientNet, OpenCV, MediaPipe.
- **Full-Stack Web & Mobile:** React.js, Next.js, Node.js, .NET Core, ASP.NET MVC, React Native, Capacitor.
- **Languages:** Python, C#, C++, JavaScript (ES6+), SQL, C.
- **Developer Tools:** SQL Server, Docker, Git, GitHub, Playwright, Trigger.dev, Entity Framework.

## LEADERSHIP

**Youth Intelligentsia (FAST Chapter)** | *President* | October 2024 – August 2025

Controlled administrative operations and team frameworks to host large-scale computing bootcamps and regional academic events.